Interpretable Machine Learning

MLR3 Tuning
Introduction & Basic Concepts

Learning goals

- Recap of hyperparameter tuning fundamentals
- Introduction to mlr3tuning package
- Basic tuning workflows and setup

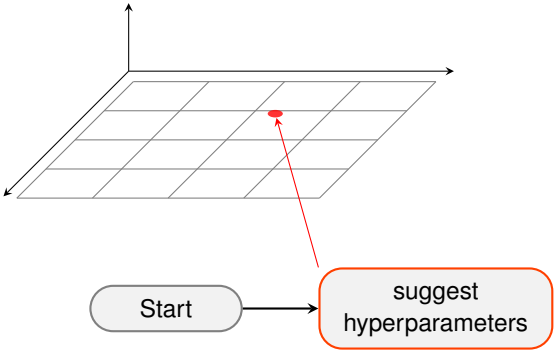
Recap: Tuning

TUNING

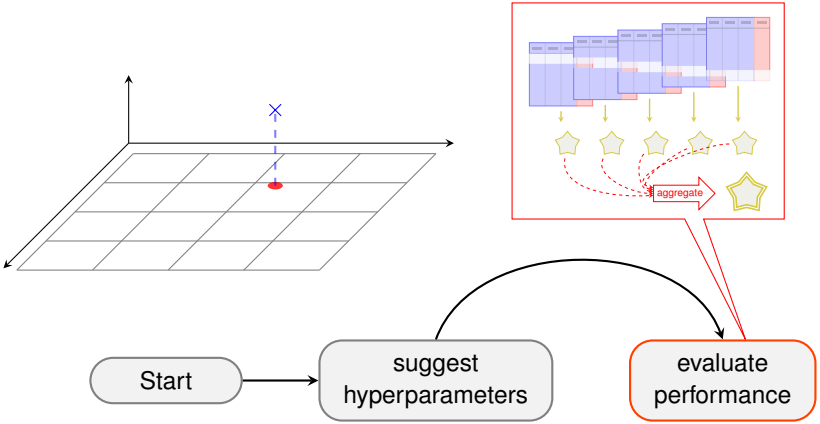
- Behavior of most methods depends on *hyperparameters*
- We want to choose them so our algorithm performs well
- Good hyperparameters are data-dependent

$\Rightarrow$ We do *black box optimization* (“Try stuff and see what works”)

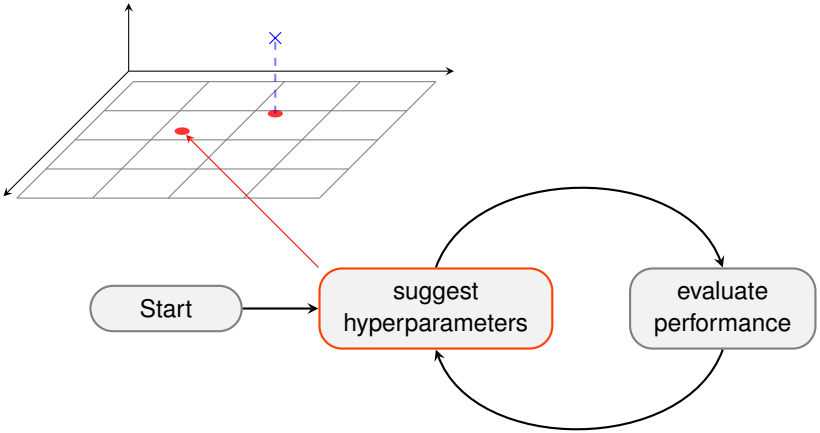
TUNING



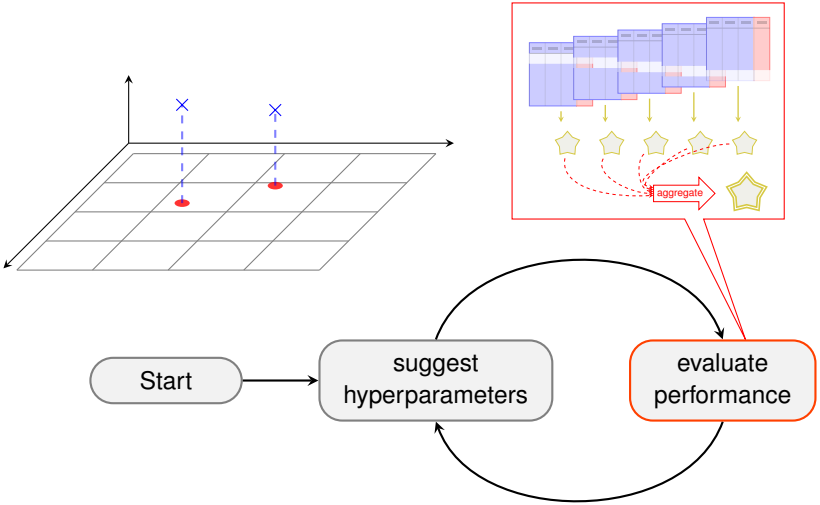
TUNING


aggregate
Start
suggest hyperparameters
evaluate performance

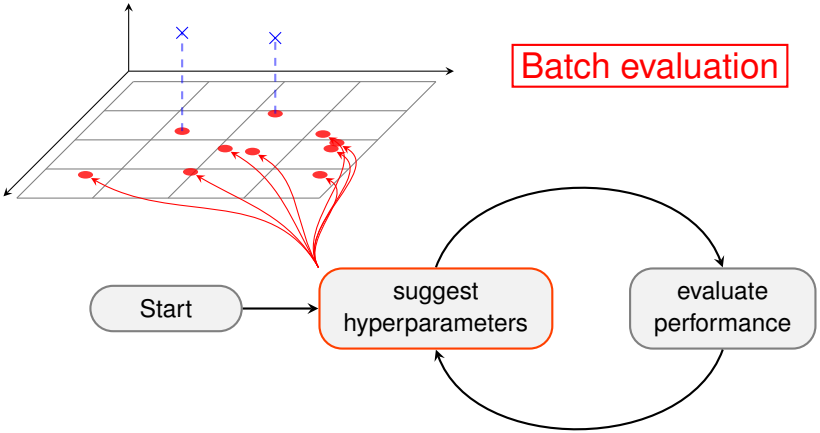
TUNING



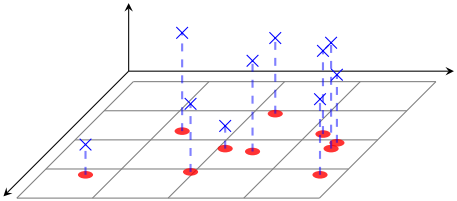
TUNING

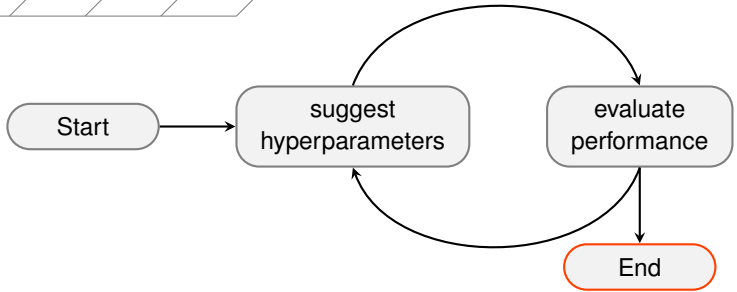

aggregate
Start
suggest hyperparameters
evaluate performance

TUNING


Batch evaluation
Start
suggest hyperparameters
evaluate performance

TUNING





Intro to mlr3tuning

TUNING

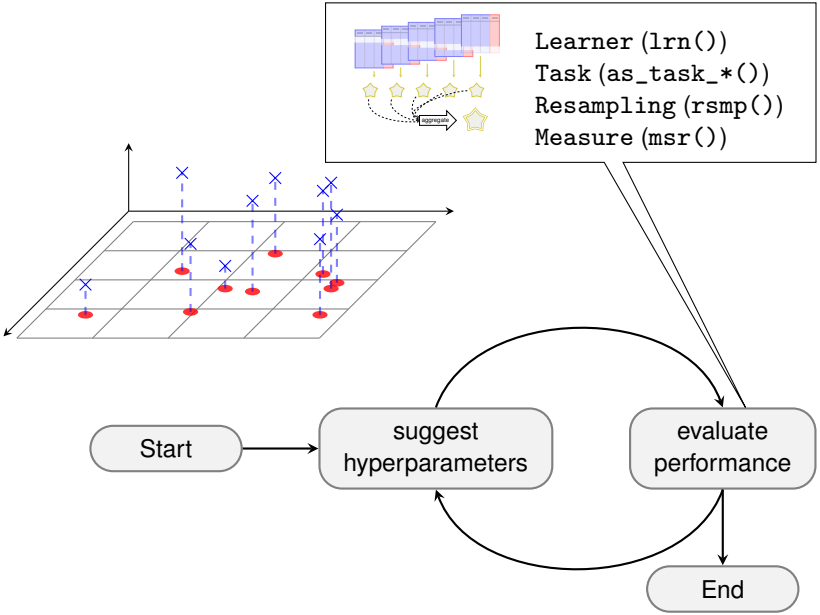
- Behavior of most methods depends on *hyperparameters*
- We want to choose them so our algorithm performs well
- Good hyperparameters are data-dependent

$\Rightarrow$ We do *black box optimization* (“Try stuff and see what works”)

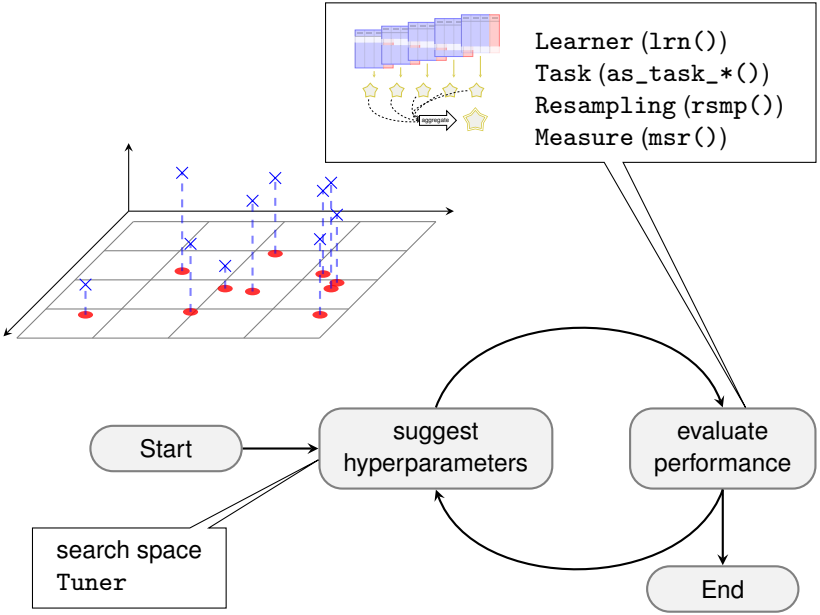
Tuning toolbox for `mlr3`:

```
library("bbotk")
library("mlr3tuning")
```

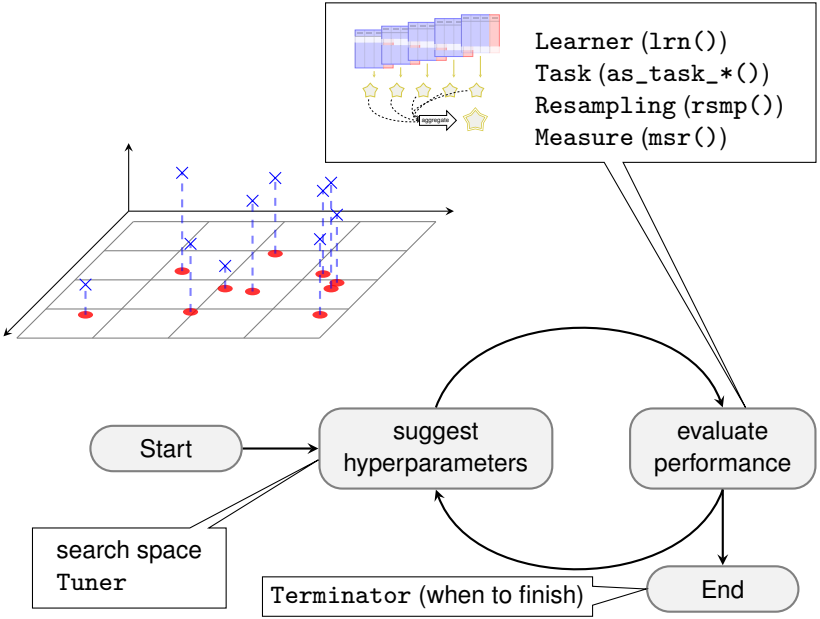
TUNING


aggregate
Learner (lrn())
Task (as_task_*())
Resampling (rsmp())
Measure (msr())
Start
suggest hyperparameters
evaluate performance
End

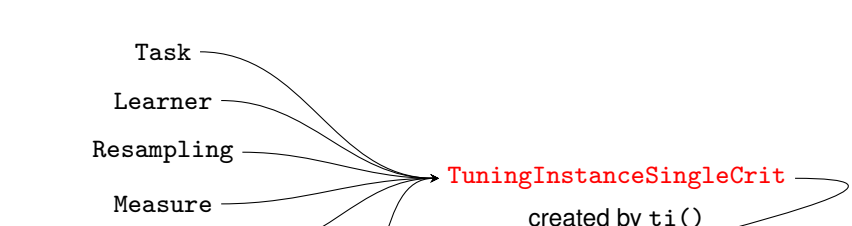
TUNING


aggregate
Learner (lrn())
Task (as_task_*())
Resampling (rsmp())
Measure (msr())
Start
suggest hyperparameters
evaluate performance
End
search space
Tuner

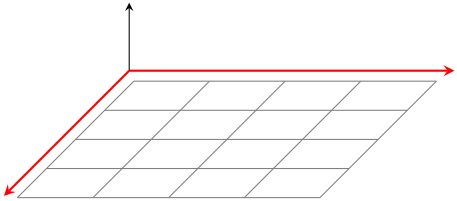
TUNING


aggregate
Learner (lrn())
Task (as_task_*())
Resampling (rsmp())
Measure (msr())
Start
suggest hyperparameters
evaluate performance
search space
Tuner
Terminator (when to finish)
End

OBJECTS IN TUNING



SEARCH SPACE SHORT FORM



```
ps(id1 = domain1, id2 = domain2, ...)
```

Numerical parameter	`p_dbl(lower, upper)`
Integer parameter	`p_int(lower, upper)`
Discrete parameter	`p_fct(levels)`
Logical parameter	`p_lgl()`
Untyped parameter	`p_uty()`

```
library("paradox")
searchspace_knn = ps(
  "k" = p_int(lower = 1, upper = 35)
)
```

SEARCH SPACE IN LEARNER DEFINITION

Specify search space for the hyperparameter to be tuned directly when creating the learner using `to_tune`:

```
library("paradox")
learner = lrn("classif.kknn", k = to_tune(lower = 1, upper = 35))
learner$param_set

#> <ParamSet>
#>             id    class lower upper nlevels default                value
#>          <char>   <char> <num> <num>   <num>  <list>               <list>
#> 1:            k ParamInt     1   Inf     Inf       7 <RangeTuneToken[2]>
#> 2:     distance ParamDbl     0   Inf     Inf       2
#> 3:       kernel ParamFct    NA    NA      10 optimal
#> 4:        scale ParamLgl    NA    NA       2    TRUE
#> 5:      ykernel ParamUty    NA    NA     Inf
#> 6: store_model ParamLgl    NA    NA       2   FALSE
```

Note: Calling `$train()` on a learner with such a specified search space does not work.

TERMINATION

- Tuning needs a *termination condition*: when to finish
- `mlr_terminators` dictionary:
-
```
as.data.table(mlr_terminators)[, c("key", "label", "properties")]
#> Key: <key>
#>                key                label             properties
#>              <char>               <char>                 <list>
#> 1:       clock_time           Clock Time single-crit,multi-crit
#> 2:            combo          Combination single-crit,multi-crit
#> 3:            evals Number of Evaluation single-crit,multi-crit
#> 4:             none                 None single-crit,multi-crit
#> 5:     perf_reached    Performance Level            single-crit
#> 6:         run_time             Run Time single-crit,multi-crit
#> 7:       stagnation           Stagnation            single-crit
#> 8: stagnation_batch     Stagnation Batch            single-crit
```

TERMINATION

- Tuning needs a *termination condition*: when to finish
- `mlr_terminators` dictionary:

```
as.data.table(mlr_terminators)[, c("key", "label", "properties")]
#> Key: <key>
#>                key                label             properties
#>              <char>               <char>                 <list>
#> 1:       clock_time           Clock Time single-crit,multi-crit
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#> 3:            evals Number of Evaluation single-crit,multi-crit
#> 4:             none                 None single-crit,multi-crit
#> 5:     perf_reached    Performance Level            single-crit
#> 6:         run_time             Run Time single-crit,multi-crit
#> 7:       stagnation           Stagnation            single-crit
#> 8: stagnation_batch     Stagnation Batch            single-crit
```

- Construct `Terminator` via `trm()` short form:

```
trm("evals", n_evals = 20)
#> <TerminatorEvals>: Number of Evaluation
#> * Parameters: n_evals=20, k=0
```

TUNING METHOD

- Choose a *tuning method* via `mlr_tuners` dictionary:

```
as.data.table(mlr_tuners)[-1, c("key", "label")]
```

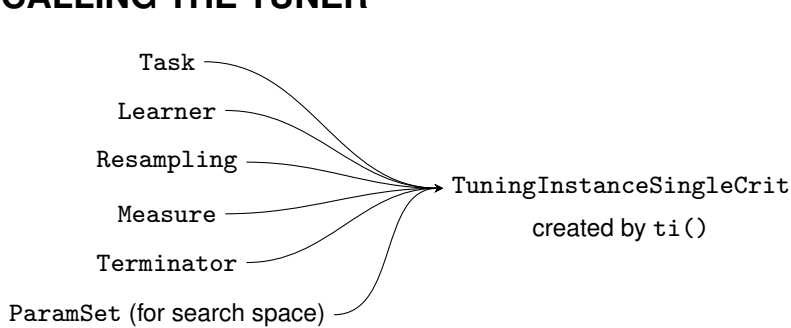
```
#> Key: <key>
#>                    key                          label
#>                 <char>                         <char>
#> 1:      design_points                   Design Points
#> 2:              gensa Generalized Simulated Annealing
#> 3:        grid_search                     Grid Search
#> 4:          hyperband                       Hyperband
#> 5:              irace                 Iterated Racing
#> 6:                mbo        Model Based Optimization
#> 7:             nloptr         Non-linear Optimization
#> 8:      random_search                   Random Search
#> 9: successive_halving               Successive Halving
```

- Construct `Tuner` via `tnr()` (`batch_size` for parallelization):

```
gsearch = tnr("grid_search", resolution = 7)
print(gsearch)
```

```
#> <TunerGridSearch>: Grid Search
#> * Parameters: resolution=7, batch_size=1
#> * Parameter classes: ParamLgl, ParamInt, ParamDbl, ParamFct
#> * Properties: dependencies, single-crit, multi-crit
#> * Packages: mlr3tuning
```

CALLING THE TUNER




```
inst = ti(task = tsk("german_credit"),
  learner = lrn("classif.kknn"),
  resampling = rsmp("holdout"),
  measure = msr("classif.ce"),
  terminator = trm("evals", n_evals = 2),
  search_space = searchspace_knn
)
```

CALLING THE TUNER

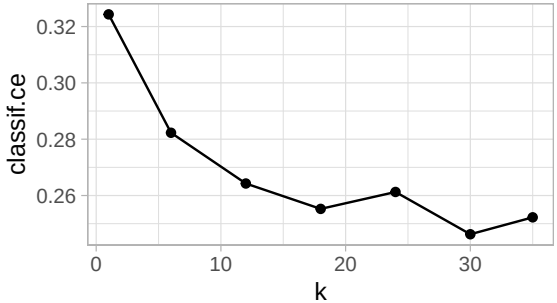
```
gsearch$optimize(inst)
#> INFO  [07:05:21.749] [bbotk] Starting to optimize 1 parameter(s) with '<TunerGridSearch>
#> INFO  [07:05:21.846] [bbotk] Evaluating 1 configuration(s)
#> INFO  [07:05:24.370] [bbotk] Result of batch 1:
#> INFO  [07:05:24.377] [bbotk]   k classif.ce warnings errors runtime_learners
#> INFO  [07:05:24.377] [bbotk]  12       0.26        0      0             0.25
#> INFO  [07:05:24.377] [bbotk]                                   uhash
#> INFO  [07:05:24.377] [bbotk]  50fda423-4a96-4c6a-ba8f-dc58fb439a73
#> INFO  [07:05:24.381] [bbotk] Evaluating 1 configuration(s)
#> INFO  [07:05:24.644] [bbotk] Result of batch 2:
#> INFO  [07:05:24.651] [bbotk]  k classif.ce warnings errors runtime_learners
#> INFO  [07:05:24.651] [bbotk]  1       0.32        0      0             0.13
#> INFO  [07:05:24.651] [bbotk]                                   uhash
#> INFO  [07:05:24.651] [bbotk]  e0d8552b-bfc4-4419-855d-0bbbe4414e4c
#> INFO  [07:05:24.665] [bbotk] Finished optimizing after 2 evaluation(s)
#> INFO  [07:05:24.667] [bbotk] Result:
#> INFO  [07:05:24.672] [bbotk]      k learner_param_vals  x_domain classif.ce
#> INFO  [07:05:24.672] [bbotk]  <int>             <list>    <list>      <num>
#> INFO  [07:05:24.672] [bbotk]     12          <list[1]> <list[1]>       0.26
#>        k learner_param_vals  x_domain classif.ce
#>    <int>             <list>    <list>      <num>
#> 1:    12          <list[1]> <list[1]>       0.26
```

TUNING RESULTS

```r
inst = ti(task = tsk("german_credit"), learner = lrn("classif.kknn"),
  resampling = rsmp("holdout"), measure = msr("classif.ce"),
  terminator = trm("none"), search_space = searchspace_knn)
gsearch = tnr("grid_search", resolution = 7)
gsearch$optimize(inst)

#>        k learner_param_vals  x_domain classif.ce
#>    <int>             <list>    <list>      <num>
#> 1:    30          <list[1]> <list[1]>       0.25

inst$archive # results are stored in tuning instance

#> <ArchiveTuning> with 7 evaluations
#>        k classif.ce batch_nr warnings errors
#>    <int>      <num>    <int>    <int>  <int>
#> 1:     6       0.28        1        0      0
#> 2:    24       0.26        2        0      0
#> 3:    18       0.26        3        0      0
#> 4:    30       0.25        4        0      0
#> 5:    12       0.26        5        0      0
#> 6:    35       0.25        6        0      0
#> 7:     1       0.32        7        0      0
```

TUNING RESULTS - VISUALIZATION

```
ggplot(as.data.table(inst$archive), aes(x = k, y = classif.ce)) +
  geom_line() + geom_point()
```



RECAP

1. Create a `Task`, `Learner`, `Resampling`, `Measure`, `Terminator` (defines when to stop), and a `ParamSet` (defines the search space):

```
task = tsk("german_credit")
learner = lrn("classif.kknn")
resampling = rsmp("holdout")
measure = msr("classif.ce")
terminator = trm("none")
searchspace_knn =  ps("k" = p_int(lower = 1, upper = 35))
```

2. Create the `TuningInstanceSingleCrit` object:

```
inst = ti(task, learner, resampling, measure,
  terminator, searchspace_knn)
```

3. Create the `Tuner` (tuning method) and optimize the learner by passing over the previously created instance to the `$optimize` method:

```
gsearch = tnr("grid_search", resolution = 7)
gsearch$optimize(inst)
```

```
#>        k learner_param_vals  x_domain classif.ce
#>    <int>             <list>    <list>      <num>
#> 1:    24          <list[1]> <list[1]>       0.24
```